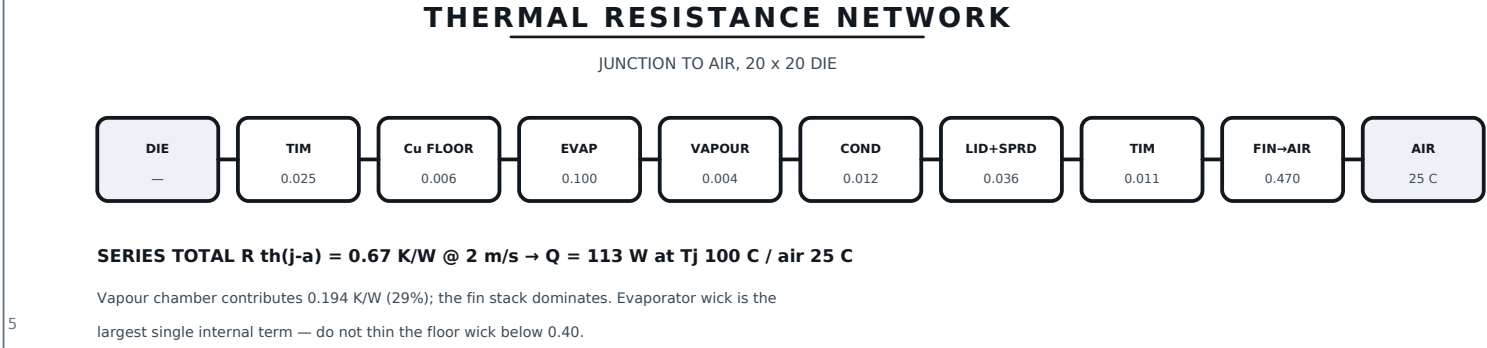
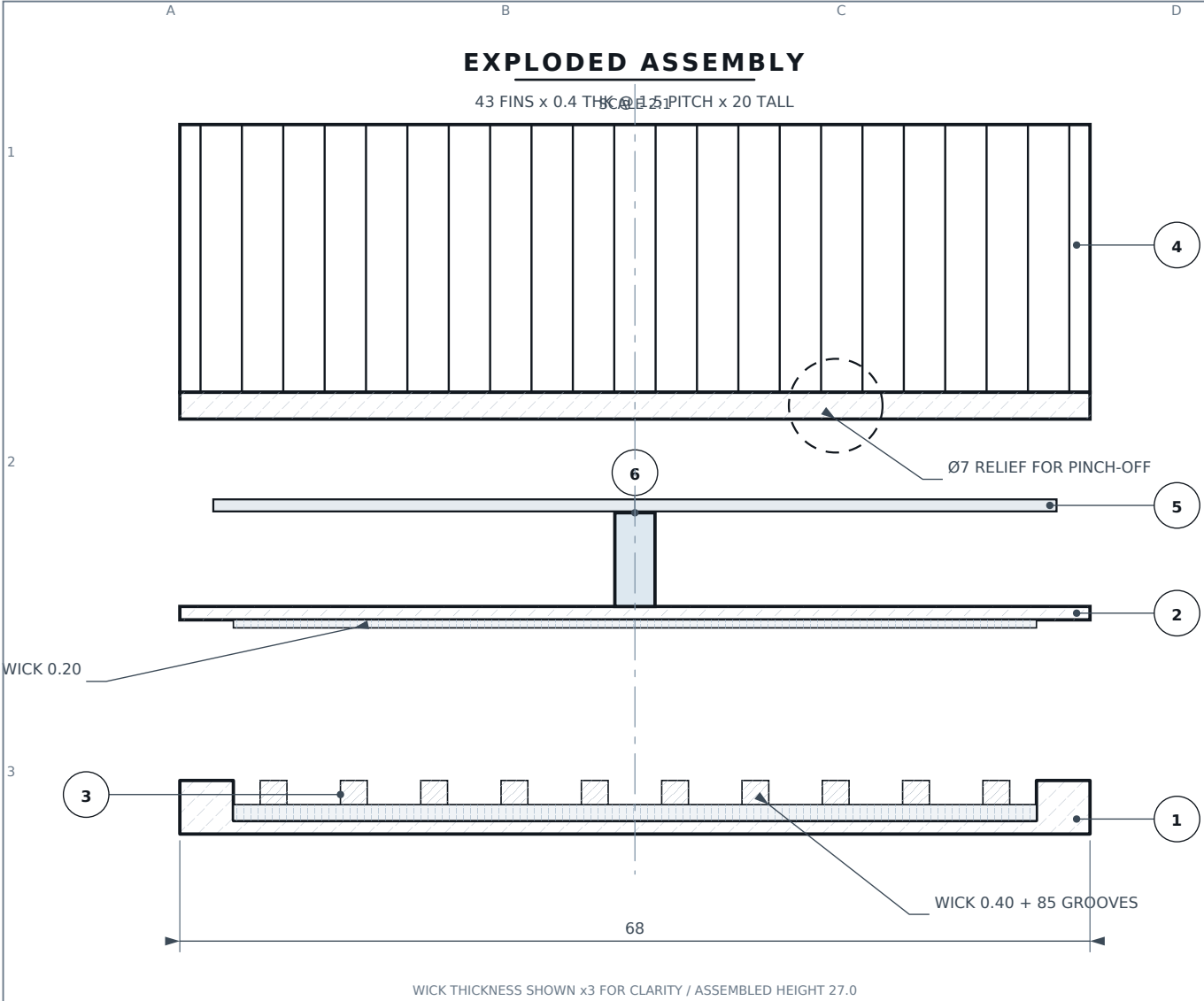


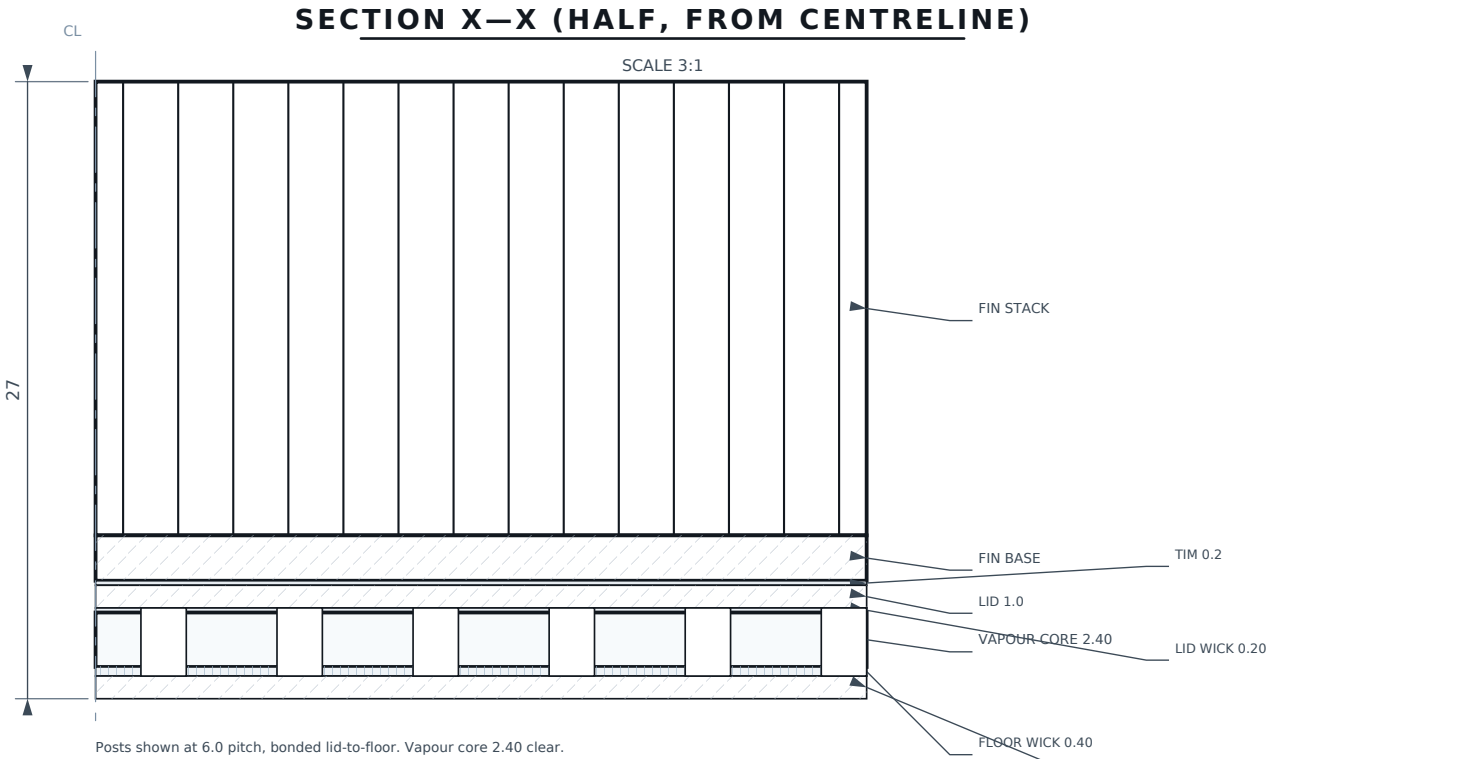


- GENERAL NOTES**
- Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
 - Charging, evacuation and sealing per VC-300. NEVER charge before diffusion bonding — bond temperature exceeds the fluid critical pressure envelope.
 - CONFIG A requires forced air $\geq 1 \text{ m/s}$. For still air use CONFIG B (VC-202); a 1.5 mm fin pitch chokes natural convection to $h = 0.03 \text{ W/m}^2\text{K}$.
 - Orientation-independent: 10.5 kPa capillary head vs 0.66 kPa gravity head over 68 mm.

BILL OF MATERIALS — ASSEMBLY VC-100					
ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	MASS
1	1	VC-101	CHAMBER BASE, POCKETED, 100 POSTS	Cu C10200 / C1020	77.4 g
2	1	VC-102	CHAMBER LID, 1.0 THK, FILL PORT	Cu C10200 / C1020	41.4 g
3	1	VC-103	SINTERED WICK — IN-SITU PROCESS	Cu POWDER 45-75um	9.6 g
4	1	VC-201	FIN STACK, 43 FIN — FORCED AIR	Al 1100-H14	88.1 g
4A	1	VC-202	FIN STACK, 13 FIN — NAT. CONVECTION	Al 1100-H14	72.7 g
5	1	VC-204	THERMAL INTERFACE PAD 0.2 THK	PCM / SIL-PAD	0.4 g
6	1	VC-205	PINCH-OFF TUBE OD3.0 x ID2.0 x 25	Cu C10200 / C1020	1.2 g
7	1.90 mL	VC-206	WORKING FLUID — DEAERATED	DI H2O 18 MOhm.cm	1.87 g

TOTAL MASS, CHARGED: 218 g (CONFIG A) / 203 g (CONFIG B)

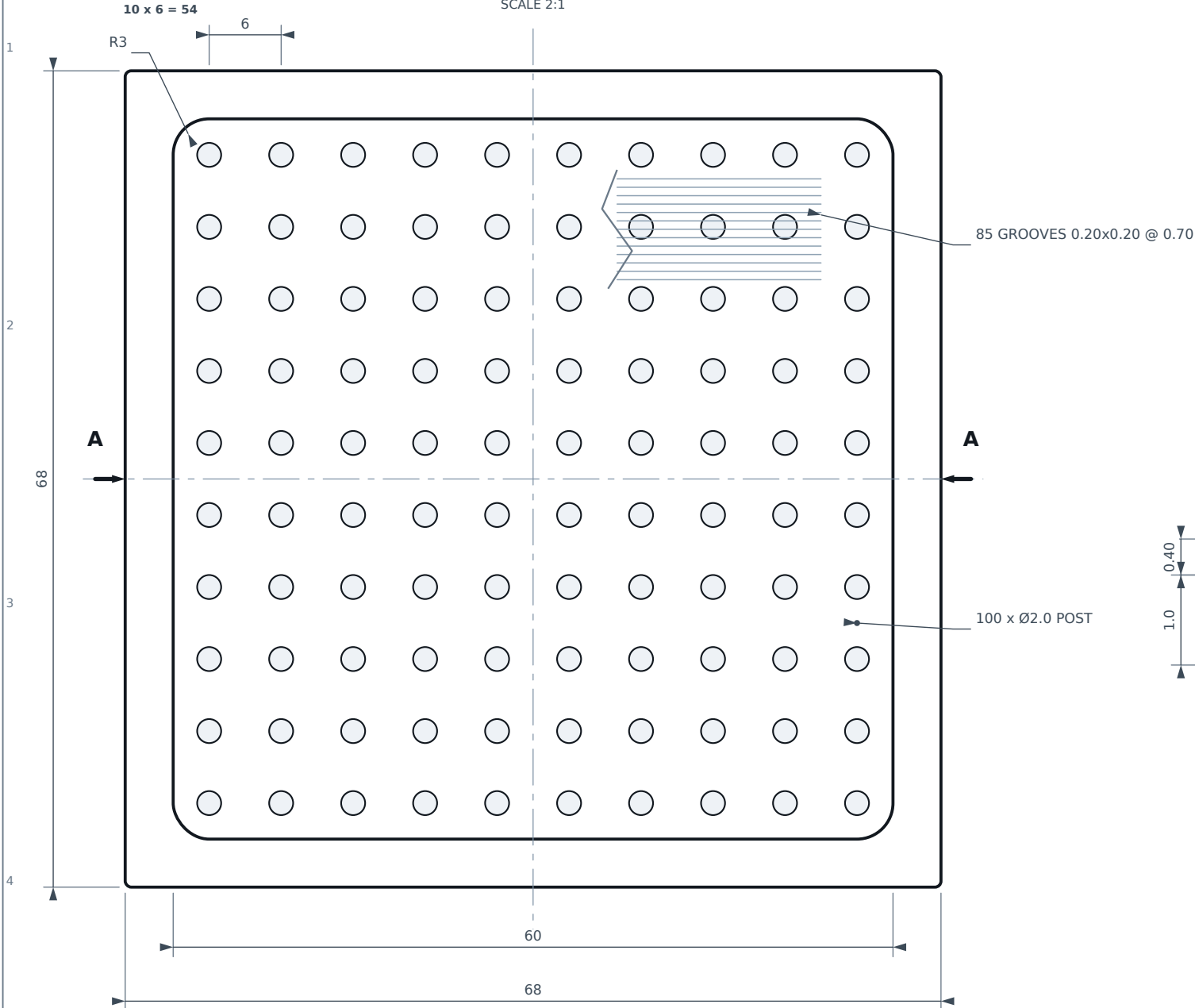
PERFORMANCE — 20 x 20 DIE, $T_j 100 \text{ C}$, AIR IN 25 C			
CONFIGURATION	AIR	$R_{th(j-a)}$	$Q @ dT 75 \text{ K}$
CONFIG A — 43 fin @ 1.5 pitch	1 m/s	1.12 K/W	67 W
CONFIG A — 43 fin @ 1.5 pitch	2 m/s	0.67 K/W	113 W
CONFIG A — 43 fin @ 1.5 pitch	4 m/s	0.46 K/W	163 W
CONFIG B — 13 fin @ 5.0 pitch	natural	3.42 K/W	22 W
CONFIG B — passive, $T_j 85 \text{ C limit}$	natural	—	16.5 W
capillary limit — water, vertical	—	—	804 W
capillary limit — water, horizontal	—	—	858 W



INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK	
TITLE GENERAL ASSEMBLY VAPOUR CHAMBER + FIN STACK / Cu-H2O / 68 x 68 x 27			SCALE 2:1		UNITS mm
			THIRD ANGLE PROJECTION		
DRAWING No. VC-100			REV A	SHEET 1 / 7	
DRAWN	CHECKED	MATERIAL	FINISH	DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09	

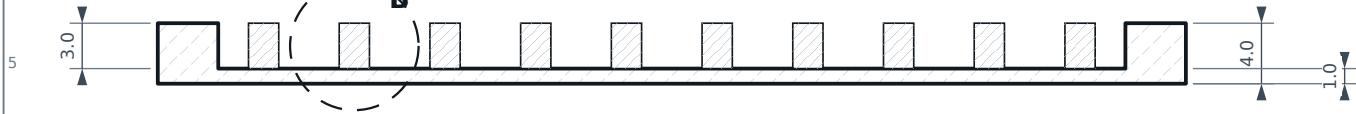
PLAN — POCKET SIDE UP

SCALE 2:1



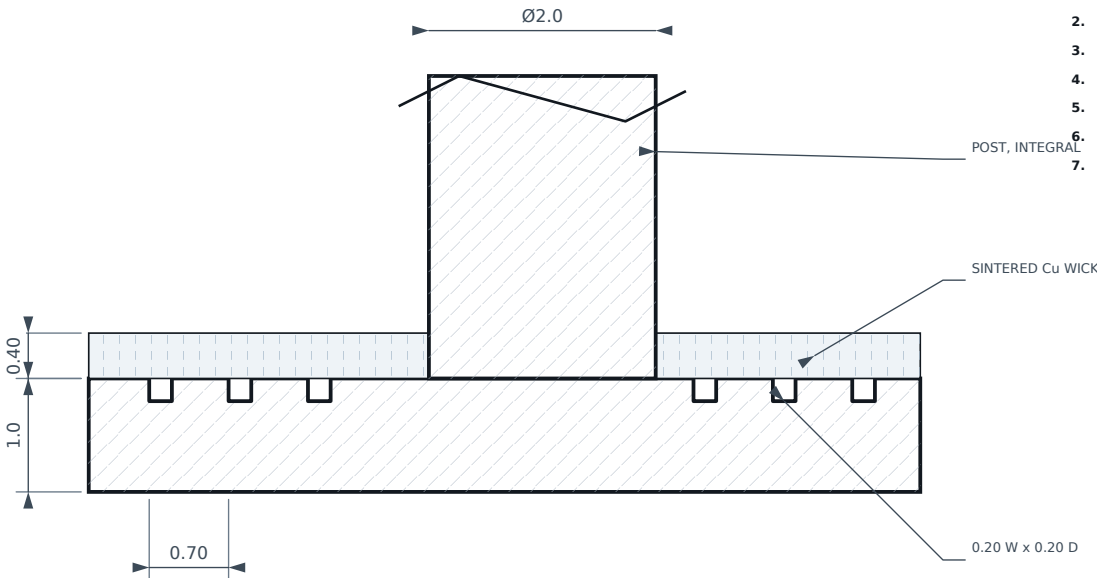
SECTION A—A

SCALE 2:1



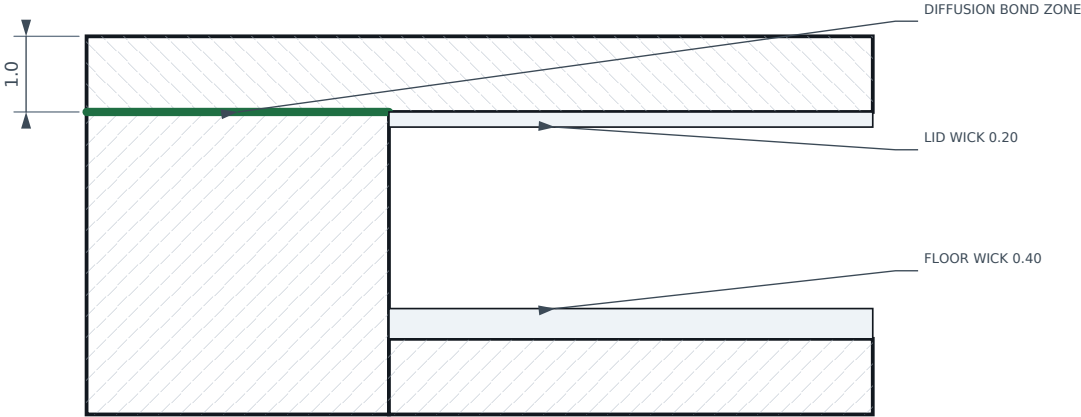
DETAIL B

SCALE 15:1



DETAIL C — RIM BOND LAND

SCALE 10:1



NOTES

- MATL: OXYGEN-FREE copper C10200 / JIS C1020, annealed. Tough-pitch copper (C11000 / JIS C1100) is PROHIBITED: it embrittles in the 950 C hydrogen sinter.
- Posts integral with base — machine from solid or coin.
- Bond faces (rim + post tops): flatness 0.02, Ra 0.4 max.
- Grooves run ONE direction only, interrupted at posts.
- Post top faces coplanar with rim within 0.015.
- Deburr 0.1 max. No burrs into cavity.
- Vacuum degrease + acid pickle before sintering.

CRITICAL CHARACTERISTICS

FEATURE	NOMINAL	TOL	CRITICAL TO
Floor thickness	1.0	+0.05 / -0.00	evaporator R th
Post height = pocket depth	3.0	±0.015	lid bond / burst
Rim + post-top flatness	—	0.02	hermeticity
Groove depth	0.20	±0.03	capillary return
Bond face roughness	Ra 0.4	max	diffusion bond
Post position	true position	Ø0.10 MMC	lid land alignment

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

CHAMBER BASE

Cu C10200 OFHC (JIS C1020) / 68 x 68 x 4.0 / 100 INTEGRAL POSTS

SCALE

2:1

UNITS

mm

THIRD ANGLE PROJECTION

DRAWING No.

VC-101

REV

A

SHEET

2 / 7

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

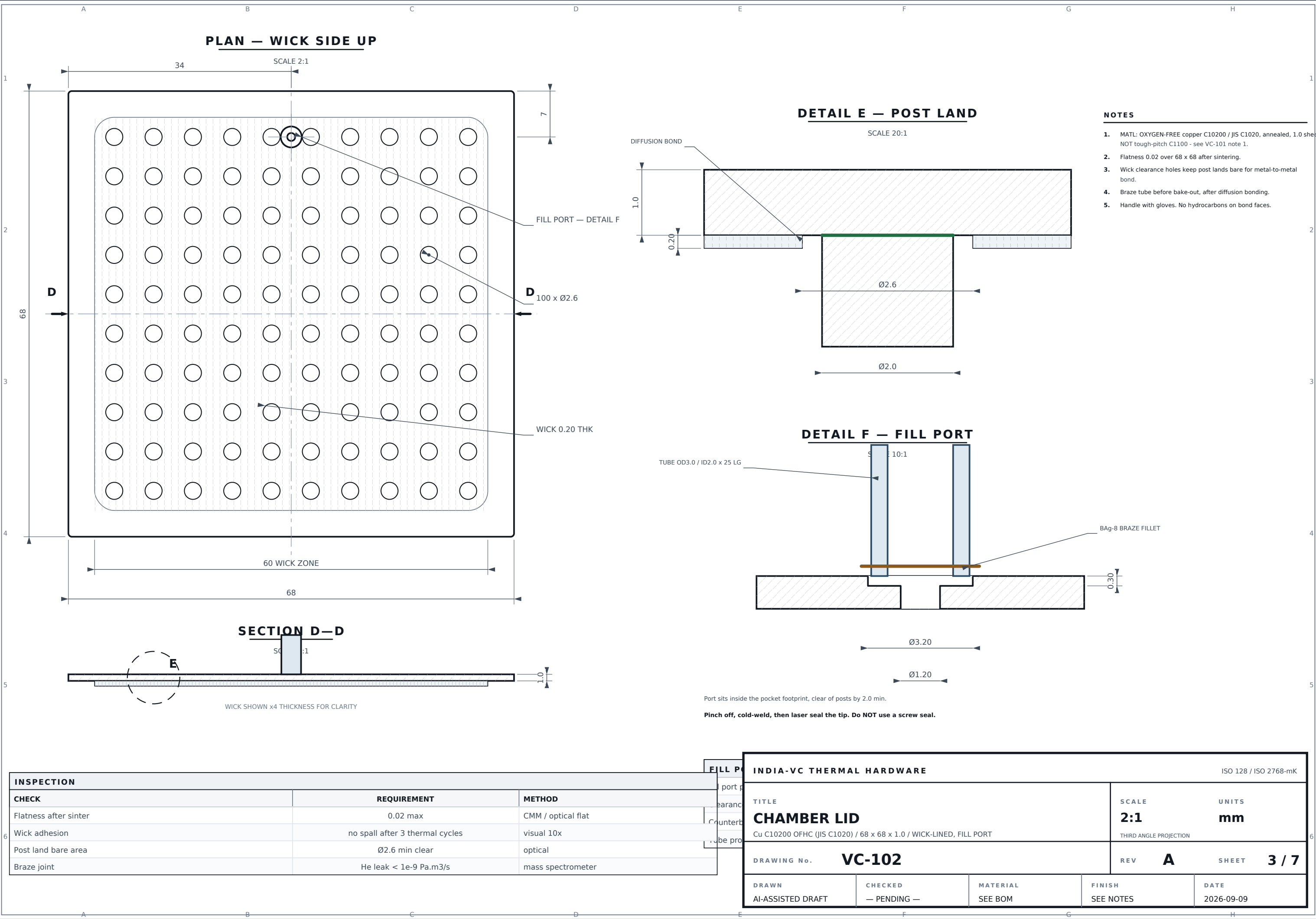
SEE BOM

FINISH

SEE NOTES

DATE

2026-09-09



INSPECTION		
CHECK	REQUIREMENT	METHOD
Flatness after sinter	0.02 max	CMM / optical flat
Wick adhesion	no spall after 3 thermal cycles	visual 10x
Post land bare area	Ø2.6 min clear	optical
Braze joint	He leak < 1e-9 Pa.m3/s	mass spectrometer

FILL PORT	INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK			
	TITLE				SCALE		UNITS		
	CHAMBER LID				2:1		mm		
	Cu C10200 OFHC (JIS C1020) / 68 x 68 x 1.0 / WICK-LINED, FILL PORT				THIRD ANGLE PROJECTION				
	DRAWING No. VC-102				REV A		SHEET 3 / 7		
DRAWN		CHECKED		MATERIAL		FINISH		DATE	
AI-ASSISTED DRAFT		— PENDING —		SEE BOM		SEE NOTES		2026-09-09	

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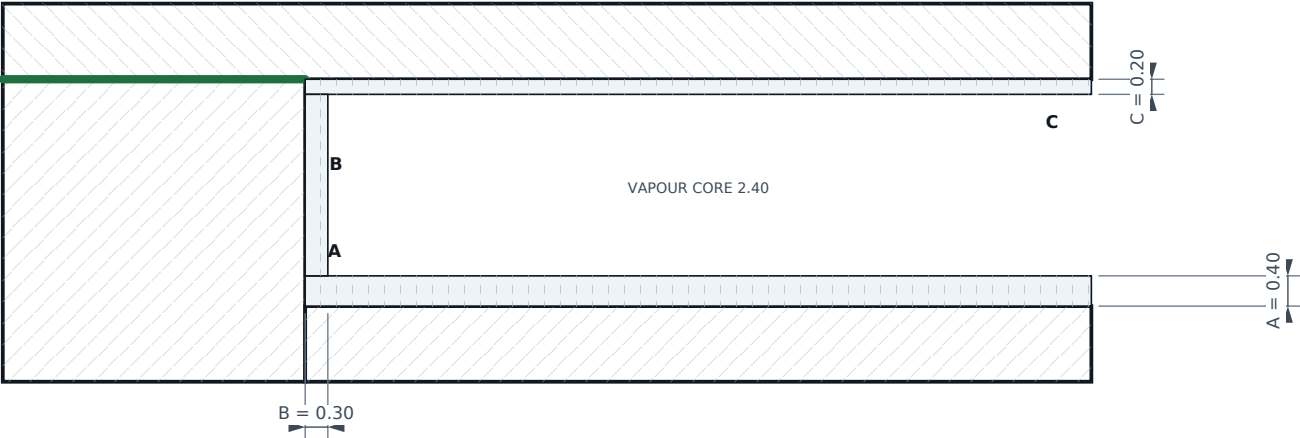
4

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6

WICK ZONE SECTION

SCALE 10:1 — ASSEMBLED



Zones A, B and C are sintered in ONE cycle and must be capillary-continuous:
condensate returns C → B → A. A break at the B/A junction causes evaporator dry-out.

WICK ZONES			
ZONE	LOCATION	THK	FUNCTION
A	Pocket floor, around posts	0.40	evaporator + liquid supply
B	Pocket side walls (4 off)	0.30	perimeter return path
C	Lid underside, posts masked	0.20	condensate collection

SINTER PROCESS		
PARAMETER	VALUE	TOL / NOTE
Cu powder, spherical, gas-atomised	45 - 75 um	sieved, -200/+325 mesh
Effective pore radius	12.6 um	= 0.21 x d(particle)
Porosity	55 %	± 5 %, quantitative metallography
Permeability (Blake-Kozeny)	1.97e-11 m2	derived, not measured
Capillary head, water @ 60 C	10.5 kPa	vs 0.66 kPa gravity head
Sintering temperature	950 C	± 10 C
Dwell	30 min	at temperature
Atmosphere	H2 (dew pt < -40 C)	or vacuum < 1e-3 Pa
Ramp / cool	10 C/min / furnace cool	no quench
Tooling	graphite mandrel + mask	masks post tops, rim, fill port

WICK ACCEPTANCE		
ACCEPTANCE TEST	REQUIREMENT	METHOD
Porosity	55 % ± 5	image analysis, 3 sections
Wick thickness A	0.40 ± 0.05	cross-section, 5 points
Capillary rise, water	> 25 mm in 60 s	witness coupon, vertical
Groove continuity	open, no blockage	back-light / CT, first article

POWDER TRADE STUDY — WHY 45-75 um				
POWDER	r eff	dP cap	Q cap VERT	VERDICT
25 um	5.2 um	25.2 kPa	348 W	clogs, low K
45 - 75 um	12.6 um	10.5 kPa	804 W	SELECTED
100 um	21.0 um	6.3 kPa	1279 W	coarse, ok
150 um	31.5 um	4.2 kPa	1802 W	marginal tilt
400 um (original drawing)	100 um	0.4 kPa	0 W	FAILS — < gravity

The original 200 um pore spec gives 380 Pa of capillary head against a 500 Pa gravity head
over 68 mm: the wick cannot lift liquid at all in any vertical orientation.

CAPILLARY PRESSURE BUDGET — WORST CASE, VERTICAL		
TERM	VALUE	NOTE
Capillary head available 2σ/r	10 508 Pa	water @ 60 C, r = 12.6 um
Liquid loss through wick + grooves	– 12.6 Pa/W	at 804 W: 10 127 Pa
Vapour loss through 2.40 core	– 0.2 Pa/W	negligible
Gravity head, 68 mm adverse	– 656 Pa	vertical, evaporator up
Margin at 113 W design point	+ 8 428 Pa	7.4 x

The wick is sized by the vertical case. Horizontal operation carries 858 W and is never the limit.

PROCESS NOTES

- Mask post top faces and the 4.0 rim before sintering — sintered powder on a bond land destroys hermeticity.
- Mask the fill port bore. Powder in the pinch-off tube prevents a gas-tight cold weld.
- Grooves must remain open under the wick. Verify by back-light or CT on the first article.
- Sinter and diffusion bond may share one furnace cycle only if the bond fixture applies load after the sinter dwell; otherwise run separately.
- No organic binders. Any residue outgasses into the sealed chamber as non-condensable gas.

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK		
TITLE SINTERED WICK — PROCESS SPECIFICATION IN-SITU SINTERED Cu POWDER / ALL WICK ZONES				SCALE 10:1		UNITS mm	
				THIRD ANGLE PROJECTION			
DRAWING No. VC-103				REV	A	SHEET	4 / 7
DRAWN		CHECKED		MATERIAL		FINISH	
AI-ASSISTED DRAFT		— PENDING —		SEE BOM		SEE NOTES	
						DATE 2026-09-09	

1

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4

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2026-09-09

EVERY STEP IN ORDER — NO SUBSTITUTION

Working fluid

Charging before diffusion bonding. Water boils at 100 C and the bond runs at 850 C: the chamber ruptures. This error appears on the superseded Maxsorb drawing — do not repeat it.

Threaded or O-ring fill seals. A screw seal cannot hold high vacuum; helium ingress accumulates as non-condensable gas and the condenser is blanketed within months. Cold-weld pinch-off only.

Aluminium anywhere wetted. Al + water generates H₂. The fin stack is external and TIM-coupled only.

Unde-aerated water. Dissolved O₂ oxidises copper and liberates H₂.

Tough-pitch copper (C11000 / JIS C1100) or brass/bronze anywhere in the envelope. Oxygen-bearing copper cracks in the hydrogen sinter; Zn and Sn evaporate under vacuum at bond temperature and contaminate the wick.

Activated carbon or any organic inside the envelope without a 250 C / 8 h vacuum bake.

Steps 7, 9 and 10 exist solely to prevent it. None of them is optional.

4 | h

reading is not traceable once the chamber is cold and under vacuum.

INDIA-VC THERMAL HARDWARE

SO 128 / ISO 2768-mK

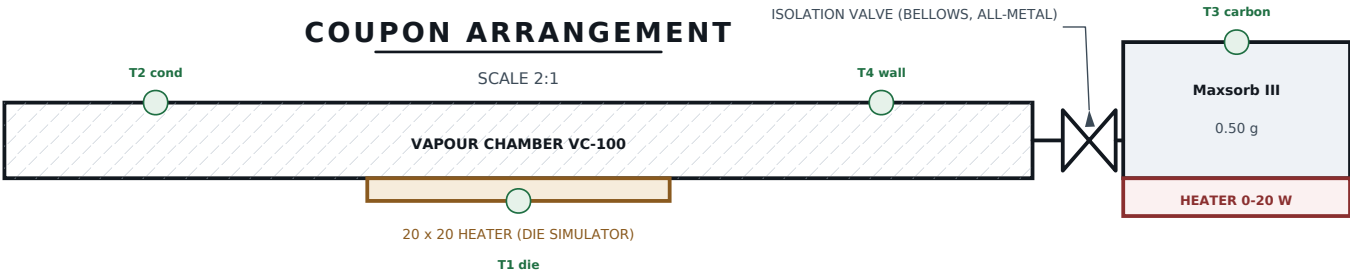
TITLE	SCALE	UNITS
PROCESS & CHARGING SPECIFICATION	NONE	mm

THIRD ANGLE PROJECTION

DRAWING No. VC-300	REV A SHEET 6 / 7
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DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-05

EXPERIMENTAL COUPON — NOT A PRODUCTION CONFIGURATION — SEE PHYSICS NOTE BELOW



Mesh: 10 um SUS 316L — NOT 50 um. Maxsorb III fines pass a 50 um mesh and foul the wick.

Carbon bake-out 250 C / 8 h / < 1e-3 Pa before installation, transferred under vacuum.

Reservoir 0.5 g Maxsorb III needs >= 1.8 cm3 at 0.28 g/cc bulk density — the original 1.50 cm3

housing could not physically hold the charge. This coupon uses 30 x 12 x 6 = 2.16 cm3.

ADSORPTION EQUILIBRIUM — DUBININ-ASTAKHOV, METHANOL / MAXSORB III						
T evap	T res	P/Psat	UPTAKE	HELD	WICK FILL	RESULT
60 C	40 C	1.00	1.31 g/g	0.66 g	15 %	DRY
60 C	60 C	1.00	1.31 g/g	0.66 g	15 %	DRY
60 C	100 C	0.24	1.06 g/g	0.53 g	35 %	DRY
60 C	127 C	0.11	0.77 g/g	0.39 g	58 %	marginal
60 C	169 C	0.03	0.42 g/g	0.21 g	90 %	wet

PHYSICS NOTE — WHY THE PASSIVE VERSION CANNOT WORK

- In a sealed pure-substance chamber the pressure is set by the evaporator: $P = P_{\text{sat}}(T_{\text{evap}})$. The carbon sees the relative pressure $P / P_{\text{sat}}(T_{\text{res}})$.
- A reservoir on the condenser side always satisfies $T_{\text{res}} \leq T_{\text{evap}}$, so $P/P_{\text{sat}} \geq 1$ and the carbon saturates: it takes 0.66 g of a 0.75 g charge and the wick runs at 15 % fill. That is permanent dry-out, not pressure control.
- Keeping the wick wet needs the reservoir ~110 K ABOVE the chip. A working vapour chamber is isothermal, so no passive layout can deliver that. It requires the heater and valve drawn above.
- Overcharging to 1.90 mL keeps the wick wet, but then liquid always remains, the pressure stays at P_{sat} , and the carbon buys nothing thermally. Its only genuine residual function is as a non-condensable gas getter — for which tens of mg suffices, not 0.5 g.

COUPON vs PRODUCTION VC-100		
	WITH CARBON	WITHOUT
Wick fill at equilibrium	15 %	125 %
Q at 2 m/s forced air	0 W (dry)	113 W
Transient buffer	722 J (heater needed)	0 J
Parts count / failure modes	+ valve, heater, control	baseline

INSTRUMENTATION		
MEASUREMENT	INSTRUMENT	PURPOSE
T1..T4	type-T TC, ± 0.2 K	gradient map
Chamber pressure	capacitance manometer	verify $P_{\text{sat}}(T)$
Reservoir mass	in-situ load cell ± 1 mg	uptake vs time
Heater power	4-wire, ± 0.5 %	energy balance
Valve state	open / closed	isolate the carbon

TEST MATRIX		
#	TEST	MEASURES
1	Valve CLOSED, sweep 0-150 W	baseline R th, no carbon
2	Valve OPEN, reservoir unheated	dry-out rate + time constant
3	Valve OPEN, reservoir held 110 K above die	whether the wick recovers
4	Valve OPEN, cyclic 10 W bursts, heater regen	usable buffer energy (722 J predicted)
5	1000 h hold, valve OPEN, unheated	NCG getter benefit vs dry-out cost

COUPON NOTES

- Fluid is METHANOL for this coupon (activated carbon is hydrophobic; water uptake is poor). Accept the 7.3x lower merit number — this coupon measures adsorption, not peak cooling.
- Methanol charge 1.90 mL so the wick stays wet with the carbon saturated. Do not run at 1.00 mL.
- The valve is what makes this an experiment rather than a defect. A production unit has no valve, so it has no way to stop the carbon draining the wick.

IF TEST 4 SUCCEEDS — PATH TO A PRODUCT

- Move the reservoir onto the die's own heat path, upstream of the chamber, so it is genuinely hotter than the evaporator without a valve.
- Replace the bellows valve with a thermally actuated one-way element, or accept a control loop driving the regeneration heater.
- Budget 722 J against ~5 g of added mass and a new hermetic joint. Compare against simply adding 15 K of copper (+50 g).

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK	
TITLE		SCALE		UNITS	
MAXSORB III RESERVOIR — TEST COUPON		2:1		mm	
EXPERIMENTAL / NOT FOR PRODUCTION / INSTRUMENTED		THIRD ANGLE PROJECTION			
DRAWING No.		VC-400		REV	A
				SHEET	7 / 7
DRAWN	CHECKED	MATERIAL	FINISH	DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09	